QXU5030 Composite Materials

2025B | Prof. Shafique Ahmed

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1. Introduction to Module

■ Assessment

Activity	Score
Coursework(labs)	40%
Final Exam	60%

Week 2N+1: Kahoot! quiz with immediate flash feedback provided after each question.

Week 2N: Quiz delivered via QM+ (or paper-based assignment), with structured written feedback provided within one week.

1.1. Basic Concepts

Anisotropy

A property of a material show different mechanical or physical properties in different directions.

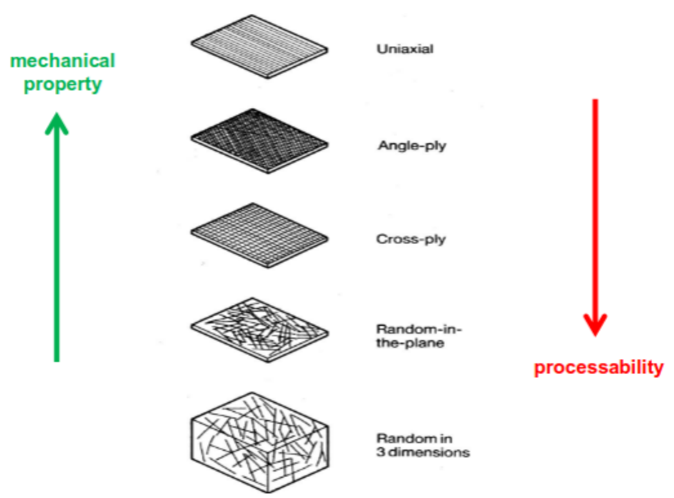
■ Man-Made Composite

Polymer Matrix Composites (PMC's) These are the most common and will be the main area of discussion in this course. Also known as **FRP** - Fibre Reinforced Polymers (or Plastics) - these materials use a polymer-based resin as the matrix, and a variety of fibres such as glass, carbon and aramid as the reinforcement.

Metal Matrix Composites (MMC's) - Increasingly found in the automotive industry, these materials use a metal such as aluminium as the matrix, and reinforce it with fibres such as silicon carbide.

Ceramic Matrix Composites (CMC's) - Used in very high temperature environments, these materials use a ceramic as the matrix and reinforce it with short fibres, or whiskers such as those made from silicon carbide and boron nitride.

■ Type of Composites



1.2. Properties of Composites

■ Factors of Composites Properties

The properties of the composite are determined by:

1. The properties of the fibre
2. The properties of the resin
3. The ratio of fibre to resin (Fibre Volume Fraction)
4. The geometry and orientation of the fibres

■ Specific Strength

$$\texttt{Strength-to-weight ratio} = \texttt{tensile strength/density}$$

Indicates how strong a material is relative to its weight.

■ Lightness, or E/ρ^3 and Lightweight Structural Design

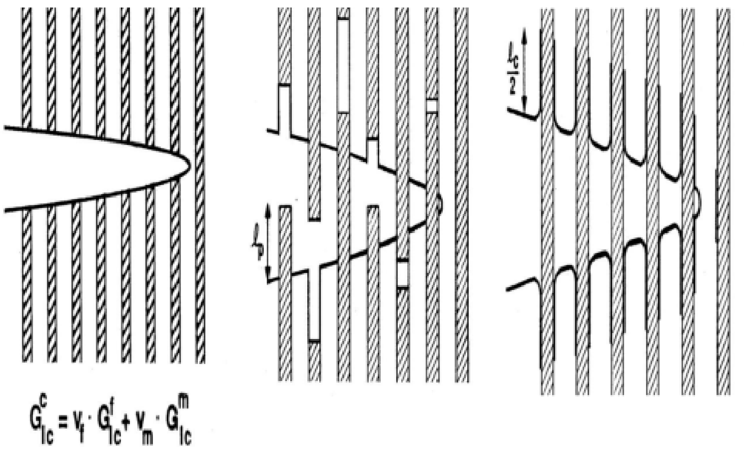
E = Young's modulus (stiffness)
E/ρ^3 = key parameter for bending stiffness and buckling resistance in weight-sensitive structures

■ Synergism, the "Composite Effect"

Ideally, rule of mixture (RoM) for a physical property P:

$$P_{\texttt{RoM}} = P_1 V_1 + P_2 V_2 \; (V_1 + V_2 = 1)$$

However, synergism can make $P_{\texttt{observed}} >> P_{\texttt{RoM}}$
And synergism can modify the mode of fraction by weakrening of fibre/matrix interface to prevent brittle fraction.



■ Safety

Composites can absorb more energy per kilo than metals in crash.

2. Fibre Reinforcements

2.1. Introduction

Most fibre have higher mechanical properties than unreinforced resin or matrix.
The four main factors that govern the fibre's contribution are:

1. The basic mechanical properties of the fibre itself.
2. The surface interaction of fibre and resin (the 'interface').
3. The amount of fibre in the composite ('Fibre Volume Fraction').
4. The orientation of the fibres in the composite.

■ Important Synthetic Fibres

Carbon - the elite fibre : stiff(higher than steel), strong, epoxies, good temperature resistance, chemically inert.

Aramid - very strong, but limited to in stiffness relative to carbon.

Glass - strong, similar to aluminium in stiffness, relatively cheap.

2.2. Glass Fibre

■ Glass Fibre Production

Blending quarry products (sand, kaolin, limestone, colemanite) at 1600°C to form liquid glass.

The liquid is passed through micro-fine bushings and simultaneously cooled to produce glass fibre filaments from 8-15 microns in diameter.

The filaments are drawn at high speed (up to 4000 m/min) together into a strand or roving.

Then, sizing and form yarns.

■ Glass Fibre Types

E-glass (electrical) - lower alkali content and stronger than A glass (alkali). Good tensile and compressive strength and stiffness, good electrical properties and relatively low cost, but impact resistance relatively poor.

C-glass (chemical) - best resistance to chemical attack. Mainly used in the form of surface tissue in the outer layer of laminates used in chemical and water pipes and tanks.

R or S-glass - higher tensile strength and modulus than E-glass, with better wet strength retention. Developed for aerospace and defence industries, and used in some hard ballistic armour applications.

E-Glass Fibre Forms

Strand. A compactly bundle of filaments, less used.
Yarns. A twisted bundle of filaments, widely used.
Roving. A untwisted, loosely associated bundle.

■ Binders and Sizing

The sizing can:

1. To protect the fibre surface from damage

2. To bind the fibres together in a yarn for ease of processing
3. To impart anti-static properties
4. To provide a chemical link between fibre surface and matrix

2.3. Carbon Fibre

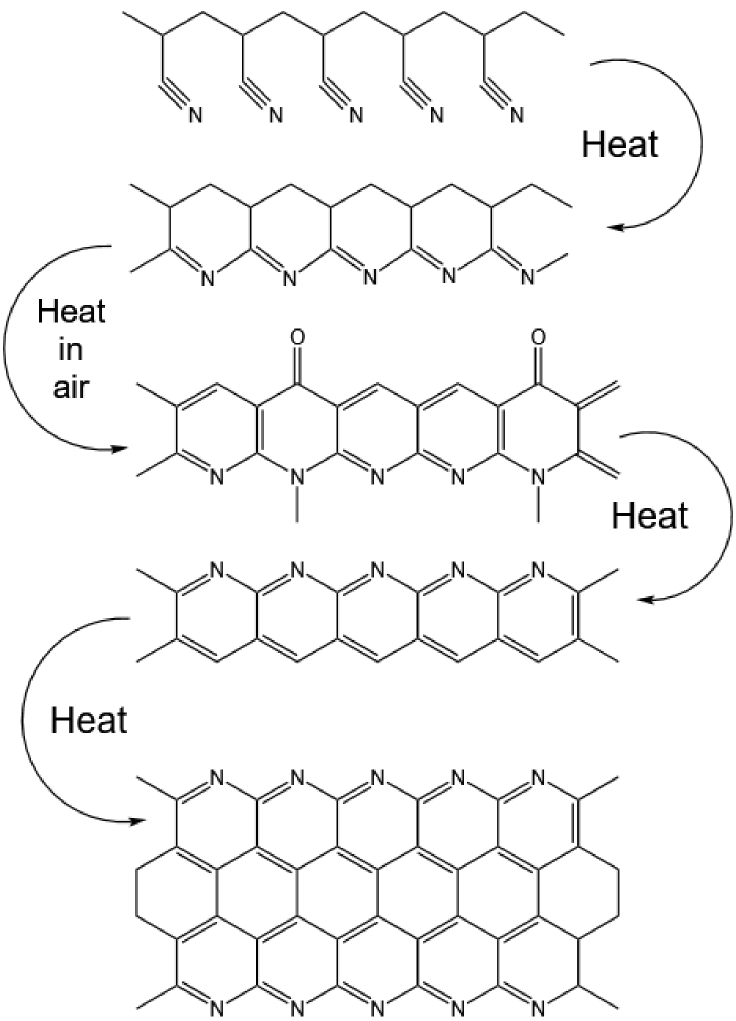
Carbon fibre is produced by the controlled oxidation, carbonisation and graphitisation of carbon-rich organic precursors(most common one is PAN) which are already in fibre form.

Carbon fibres can be grouped as:

1. High strength (HS)
2. Intermediate modulus (IM)
3. High modulus(HM)
4. Ultra high modulus(UHM)

Carbon fibre has the highest specific stiffness of any commercially available fibre, in both tension and compression (in case of HS-fibre) and a high resistance to corrosion, creep and fatigue.

Their impact strength, however, is lower than either glass or aramid, particularly HM and UHM fibres.

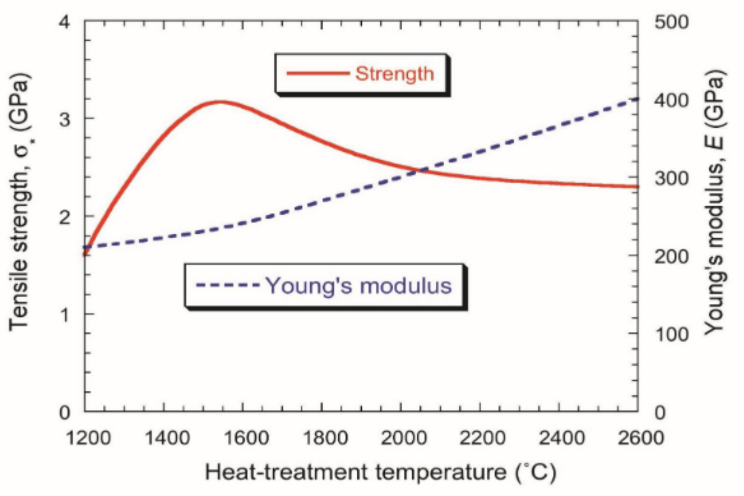


■ Carbon Fibre Production by PAN

Tensile modulus of CF can mainly be determined by carbonization/graphitization temperature. Also, the nature of precursors (PAN, pitch) affects on the easiness of reaching a certain level modulus.

Some clothes says ”acrylic”, is a blended copolymer of polyacrylonitrile, with low purity and short chain.

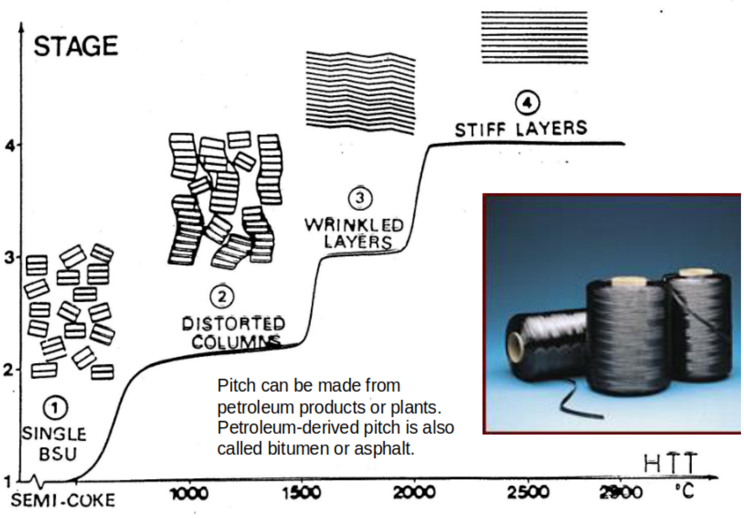
Here we show the effect of heat treatment on strength and Young’s modulus.



■ Carbon Fibre Production by Mesophase Pitch

Heating to

Single BSU → Distorted Columns → Wrinkled Layers → Stiff Layers



Graphitisation between 1500 and 3000°C gives improved orientation and properties. Heating above 2000°C gives higher crystallinity and higher (anisotropic) modulus - HM-Carbon.

2.4. Aramid Fibres

Aramid fibre is a man-made organic polymer produced by spinning from a liquid crystalline solution.

1. High strength and low density.
2. Good resistance to impact, abrasion, chemical and thermal degradation.
3. Low elongation to break.
4. Low electrical conductivity.
5. **POOR** compressive strength, and degradate under UV light.

Aramid fibres are highly anisotropic - 1D - with strong covalent bonds in the fibre direction and weak hydrogen bonding between the chains.

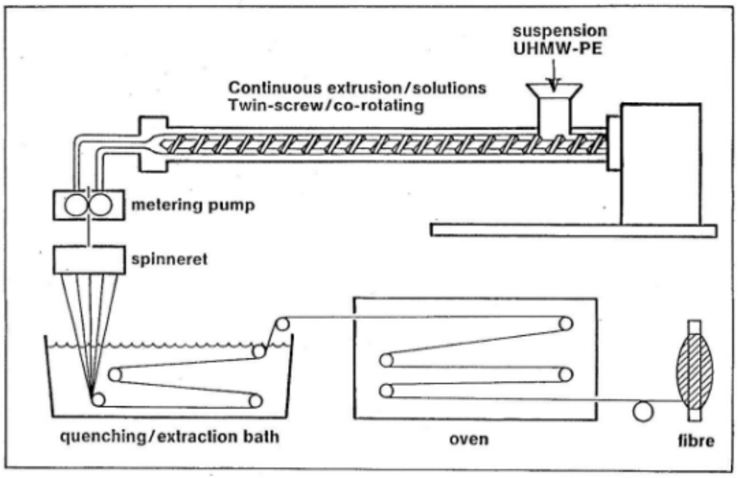
This highly anisotropic character results in a low shear modulus and weak transversal and compressive properties.

2.5. Polyethylene Fibres

High-performance polyethylene fibres - like Dyneema or Spectra - are the world's strongest and lightest fibres.

It has outstanding toughness, low density and high resistance to chemicals, water, and ultraviolet light, and can absorb extremely high amounts of energy.(To use these properties, a unidirectional sheet called Dyneema UD or Spectra Shield was developed)

It has excellent vibration damping, flex fatigue and internal fiber-friction characteristics, and low dielectric constant makes it virtually transparent to radar.



Solution (gel) spinning process. The highly drawn fibre contains an almost 100% crystalline structure with nearly perfectly arranged molecules, which promotes its extremely high strength, modulus, and other excellent properties.

Free breaking length

This is the theoretical length at which a fiber will break under its own weight.
The free breaking length is independent of the size of the fiber and is determined only by the material properties. HPPE has higher free breaking length than polyester or polyamide.

2.6. Boron Fibres

Produced by **Chemical Vapour Depostion** on tungsten or carbon wire. Advantages of boron fibres are:

1. High modulus (400 GPa)
2. High strength (3500 MPa)
3. High temperature resistance (700 °C)
4. high compressive strength.

However, it is so expensive that restrict its use.

2.7. Ceramic Fibres

Silicon carbide fibres are also made by CVD using a tungsten wire as substrate. However usually they are in the form of very short ’whiskers’ (Nicalon) and are mainly used in areas requiring high temperature resistance.

They are more frequently associated with non-polymer matrices such as metal alloys.

2.8. Natural Fibres

Natural fibres (flax, jute, hemp, sisal) are renewable, cheap (€0.35-1.5/kg), lightweight, and biodegradable.

Advantages and Disadvantages vs Glass

Advantages:
Low density (1.3-1.5 g/cm^3), no skin irritation, non-abrasive to equipment, CO_2 neutral.
Disadvantages:
Moisture sensitive, limited heat resistance, irreproducible quality, poor impact and compressive strength (fail by kink-band formation).

Property	Flax	Jute	Sisal	Glass
Density (g/cm^3)	1.54	1.46	1.33	2.55
Strength (MPa)	600-1500	400-800	600-700	2400
Stiffness (GPa)	40-70	10-30	38	73
Price (€/kg)	0.5-1.5	0.35	0.6-0.7	1.3

Applications include press moulding parts and wood fibre plastics (”mouldable wood”).

2.9. Textile Constructions

■ Fibre Finishes

To allow handling with minimum damage and to promote fibre/matrix interfacial bond strength.

Glass Fibre Finishes: ’Dual-function’ finish for direct fibre processes (prepregging, pultrusion, filament winding). For weaving: protective finish (starch-based) applied first, then scoured and treated with matrix-compatible finish.

Carbon Fibre Finishes: Epoxy based, 1-2% for weaving, 0.5-1% for tape prepregging or filament winding.

Aramid Fibre Finishes: Composite finish, rubber compatible finish (belts and tyres), waterproof finish (ballistic soft armour).

Non-woven Mat

Chopped strand mat (CSM) and continuous strand mat in non-woven, random pattern for equal distribution in all directions.

- **Chopped strand mat**: Strands chopped to various lengths, deposited randomly. Used for hand lay-up, continuous laminating.
- **Continuous strand mat**: Deposited in swirl pattern. Used for press molding, RTM, pultrusion.

Available in weights from 200 to 1000 g/m^2.

Woven Fabrics

Woven roving: Heavy, coarse, drapable fabric of plain weave. Alternated with CSM in multilayer laminates to increase in-plane shear and peel strength.

Weave styles:

- **Plain weave**: Each warp fibre passes alternately under and over each weft fibre. Symmetrical, good stability, most difficult to drape, high crimp lowers mechanical properties.
- **Twill weave**: One or more warp fibres weave over and under two or more weft fibres. Superior wet out and drape, smoother surface, slightly higher mechanical properties.
- **Satin weave**: Fewer intersections of warp and weft (harness number 4, 5, 8). Very flat, good wet out, high drape, low crimp gives good mechanical properties.

Hybrid Fabrics

Fabric with more than one type of structural fibre.

- **Carbon/Aramid**: High impact resistance and tensile strength of aramid + high compressive stiffness of carbon. Both low density but high cost.
- **Aramid/Glass**: Low density, high impact resistance of aramid + good compressive and tensile strength of glass, lower cost.
- **Carbon/Glass**: High tensile/compressive stiffness of carbon + reduced density, glass reduces cost.

Unidirectional Fabrics

Majority of fibres runs in one direction only ($\sim$10% fibre in other direction to hold primary fibres in place).

UD prepreg: True UD fabrics offer highest possible fibre properties. Fibres are straight and uncrimped. Only resin system holds fibres in place.

Multi-axial Stitched Fabrics

One or more layers of long fibres held in place by PET stitching thread.

Advantages: Fibres always straight and non-crimped; fewer layers needed; better wet-out and permeability.

Disadvantages: PET stitching doesn't bond well to some resins; slow production process; high machinery and fabric cost.

■ Braided Fabrics

Fibres interlaced in spiral nature to form tubular fabric. Diameter controlled by number of fibres, spiral angle (25° to 75°), intersections per unit length, and fibre size.

Used for masts, antennae, drive shafts and other tubular structures requiring torsional strength.